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Mr. / Miss Harsh . Deepak . Salkar *of the*

M.Sc. Computer Science *Class Seat No.* 171 *has*

satisfactorily completed the required number of practicals for

Semester I *in the*

Course Research Methodology

Course Code RPSRMCS0505 *as per the syllabus*

during the academic year **2026- 2027**

Subject In-Charge
Date: 09/10/26

Head of Department
Date: 09/10/26

Examiner
Date:

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RAMNARAIN RUIA AUTONOMOUS COLLEGE
DEPARTMENT OF COMPUTER SCIENCE & IT
Academic Year 2026-2027

ROLL NO - 171 STUDENT NAME - Harsh.Deepak.Salkar.

CLASS: M.Sc. Comp Sci SEMESTER: I (2026-27) COURSE CODE: RPSRMCSO505

COURSE NAME - RESEARCH METHODOLOGY FACULTY IN-CHARGE - MAHAVIR ADVAYA

INDEX

SR. NO.	DATE	PARTICULAR	SIGNATURE
1	16-07-2026	Identification of a topic / research problem in Computer Science / IT	
2	23-07-2026	Conduct literature review for the selected topic using Google Scholar	
3	13-08-2026	Formulation of research objectives and hypotheses	
4	17-08-2026	Prepare annotated bibliography of 10 research papers	
5	20-08-2026	Design research methodology for the selected problem	
6	24-09-2026	Sample Data Collection and Design the Software Framework	
7	08-10-2026	Perform data analysis / develop an algorithm	
8	08-10-2026	Generate statistical summaries / charts / testing on data	

[Practical No - 1]

1.] Aim : To identify and select a feasible research problem in Blockchain Security and justify its significance by proposing an Artificial Intelligencebased solution for detecting blockchain security attacks.

A.] Area of Interest :

Blockchain technology has become one of the most important innovations in Computer Science because it enables decentralized, secure, and tamper-resistant data management. It is widely used in cryptocurrencies, banking, healthcare, supply chain management, Internet of Things (IoT), digital identity, and many other real-world applications.

Although blockchain provides strong security through cryptographic techniques and decentralized consensus mechanisms, it is still vulnerable to several cyberattacks such as **51% attacks, Sybil attacks, Selfish Mining, Double Spending, Eclipse attacks, and Smart Contract vulnerabilities**.

In recent years, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as promising technologies for detecting abnormal patterns and cyber threats automatically. AIbased security systems can analyze blockchain transaction data, identify malicious behavior, and assist in protecting blockchain networks from evolving attacks.

Because blockchain technology is increasingly being adopted in critical sectors, improving its security through Artificial Intelligence has become an important research direction.

B.] Identifying the Research Problem :

Blockchain networks are generally considered secure due to decentralization and cryptographic protection. However, attackers continuously develop sophisticated methods to exploit vulnerabilities within blockchain ecosystems.

Existing blockchain security solutions often focus on detecting only a specific type of attack or rely on static rule-based mechanisms. Many of these approaches require high computational resources, cannot operate in real time, and provide limited explanation regarding their predictions.

As blockchain adoption continues to grow, there is an increasing need for intelligent security mechanisms capable of detecting multiple attacks accurately while maintaining scalability and transparency.

This tells us:

- What is the problem?
- Why is it important?
- Why does it require further research?

So our research problem can be stated as:

Existing blockchain security systems have limited capability to detect multiple blockchain attacks in real time while maintaining high accuracy, scalability, and explainability

C.] Why Existing Solutions are Insufficient :

A review of existing blockchain security research indicates several limitations.

Most existing studies

- Detect only one type of blockchain attack.
- Focus mainly on Bitcoin or Ethereum individually.
- Use offline datasets instead of real-time blockchain transactions.
- Do not provide Explainable AI (XAI).
- Require high computational resources.
- Have limited scalability for large blockchain networks.
- Rarely compare multiple machine learning algorithms under identical experimental conditions.

These limitations indicate that current blockchain security solutions still require significant improvement.

D.] Selecting the Research Topic :

After identifying the research problem and reviewing the existing literature, the next step is selecting an appropriate research topic.

Rather than proposing another traditional blockchain security mechanism, this research aims to investigate how Artificial Intelligence and Machine Learning can improve blockchain attack detection through intelligent, explainable, and lightweight models.

Selected Research Topic

A Lightweight Explainable Machine Learning Framework for Real-Time Detection of Blockchain Security Attacks

E.] Identifying the Research Gap :

The literature review indicates that researchers have explored various blockchain security techniques, including

- Machine Learning
- Deep Learning
- Explainable AI
- Smart Contract Security
- Blockchain Anomaly Detection
- Fraud Detection
- Intrusion Detection

However, there is limited research focusing on a unified framework capable of

- Detecting multiple blockchain attacks simultaneously.
- Performing real-time attack detection.
- Supporting multiple blockchain platforms.
- Providing Explainable AI.
- Maintaining low computational overhead.

Therefore, these limitations represent an important research gap that motivates the proposed work.

F.] Research Objectives :

The objectives of this research are:

- To study common blockchain security attacks.
- To analyze existing AI and Machine Learning approaches for blockchain security.
- To compare different machine learning algorithms for blockchain attack detection.
- To develop a lightweight AI-based framework capable of detecting multiple blockchain attacks.
- To evaluate the proposed framework using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, and Detection Time.

G.] Expected Contribution :

The proposed research is expected to contribute by

- Improving blockchain attack detection accuracy.
- Supporting detection of multiple blockchain attacks within a unified framework.
- Providing explainable predictions using Explainable AI.
- Reducing computational complexity through lightweight machine learning techniques.
- Enhancing blockchain security for real-world applications.

H.] Justification of the Selected Topic :

Blockchain technology is rapidly becoming an essential component of modern digital infrastructure. Financial institutions, healthcare organizations, supply chain companies, government agencies, and decentralized applications increasingly depend on blockchain for secure data management.

As blockchain adoption expands, cyberattacks targeting blockchain systems also continue to evolve. Developing intelligent, explainable, and efficient security mechanisms has therefore become an important research challenge.

This research aims to address this challenge by investigating how Artificial Intelligence and Machine Learning can improve blockchain attack detection while maintaining high accuracy, transparency, and scalability.

Therefore, this topic represents a relevant, practical, and feasible direction for future research.

[Practical No - 2]

Literature Review

Research Topic – AI-Based Detection of Blockchain Security Attacks Using Machine Learning

Sr. no	Author(s) & Year	Focus / Study Type	Key Findings	Methodology	Research Gap Identified
1.	Vanessa Ortiz, Alexandria Bates, Gaby G. Dagher, Jun Zhuang, Tim Andersen (2025)	Survey of Deep Learning techniques for Blockchain Security	Deep learning significantly improves blockchain attack detection, fraud analysis, and anomaly detection, but adoption across blockchain layers is still limited.	Categorizes research by neural network type (CNN, RNN, LSTM, GNN, GAN, Transformers), blockchain layer, security/privacy objective, and attack/defense applications.	Need benchmark datasets, real-time deployment, scalable AI models, privacy-preserving AI, and protection for consensus and hardware layers.
2.	Xiaoqi Li, Peng Jiang, Ting Chen, Xiapu Luo, Qiaoyan Wen (2018)	Comprehensive Blockchain Security Survey	Identifies major attacks including 51% attack, Selfish Mining, Eclipse attack, Double Spending and Smart Contract vulnerabilities.	Reviews blockchain architecture, attacks, vulnerabilities, and defense mechanisms across different blockchain systems.	Future work needed in scalability, consensus security, privacy protection, smart contract verification and attack detection.
3.	Liehuang Zhu, Baokun Zheng, Meng Shen, Shui Yu, Feng Gao, Hongyu Li, Kexin Shi, Keke Gai (2018)	Blockchain Data Security Survey	Provides one of the first comprehensive taxonomies of blockchain data security threats and countermeasures.	Classifies attacks into Privacy, Integrity, Availability and Controllability categories, then analyzes corresponding defenses.	Privacy preservation, secure smart contracts, secure data sharing and blockchain data management remain open research areas.

4.	Tara Salman, Maede Zolanvari, Aiman Erbad, Raj Jain, Mohammed Samaka (2018	Blockchain-based Security Services	Blockchain provides decentralized security services but introduces performance and scalability overhead in practical deployments.	Reviews blockchain-based authentication, confidentiality, integrity, provenance, access control and privacy mechanisms.	Need lightweight blockchain frameworks, interoperability between platforms, improved scalability and lower latency.
5.	Gheyath Mustafa Zebari, Noor Al Musalhi (2025)	AI + Blockchain Security Review	AI enhances blockchain security through intelligent threat detection and automated decision-making.	Reviews AI integration into blockchain security for intrusion detection, fraud detection, smart contracts and cybersecurity.	Need Explainable AI (XAI), standardized evaluation methods, lightweight AI models, and real-world validation.
6.	Vasavi Chithanuru, Mangayarkarasi Ramaiah (2023)	AI-based Blockchain Anomaly Detection Review	AI significantly improves anomaly detection, fraud detection, and malicious node identification across blockchain environments.	Reviews Machine Learning, Deep Learning, Graph Learning, and Explainable AI techniques used for blockchain anomaly detection.	Lack of benchmark datasets, real-time deployment, lightweight AI models, and explainable detection systems.
7.	Mohammad Hasan, Mohammad Shahriar Rahman, Helge Janicke, Iqbal H. Sarker (2024)	Blockchain Transaction Anomaly Detection	Ensemble models outperform individual classifiers, and SHAP improves interpretability of anomaly detection	Uses Random Forest, XGBoost, Voting Ensemble, Stacking Ensemble, SHAP Explainability, and introduces XGBCLUS balancing algorithm.	More generalized datasets, real-time streaming detection, and cross-blockchain evaluation are still needed.
8.	Yeajun Kang, Wonwoong Kim, Hyunji Kim, Minwoo Lee, Minho Song, Hwajeong Seo (2023)	Smart Contract Malware Detection	Achieved **92.3% detection accuracy** while reducing model size by **41.5%**, making it suitable for IoT environments.	Developed a lightweight deep learning model (Gredeeptector) with Explainable AI (XAI) to detect malicious smart contracts.	Needs testing on more blockchain platforms and additional attack categories beyond greedy contracts.

9.	Beibei Han, Yingmei Wei, Qingyong Wang, Francesco Maria De Collibus, Claudio J. Tessone et al. (2024)	Graph Neural Network for Ethereum Fraud Detection	GNN captures temporal and structural relationships better than traditional ML methods, improving anomaly detection accuracy.	Uses Graph Neural Networks (GNN) with temporal transaction modeling for anomaly detection.	High computational cost and limited scalability for very large blockchain networks remain challenges.
10.	Prabhat Kumar, Danish Javeed, Randhir Kumar, A.K.M. Najmul Islam (2024)	Blockchain + Explainable AI	Explainable AI increases transparency, improves analyst confidence, and supports secure blockchain-based threat detection.	Integrates Blockchain with Explainable AI to improve cybersecurity decision-making and trust in AI predictions.	Future work includes developing lightweight XAI frameworks and testing on real blockchain transaction datasets.
11.	Kaustubh Dwivedi, Rajat K. Pal, et al. (2024)	Blockchain Attack Classification	Provides a comprehensive attack classification and maps existing defense mechanisms.	Proposes a layer-wise taxonomy of blockchain attacks covering Network, Consensus, Data, Smart Contract, and Application layers.	Lacks automated AI-based multi-layer attack detection and real-time mitigation.
12.	Safak Kayikci & Taghi M. Khoshgoftaar (2024)	Blockchain + Machine Learning Survey	Shows ML significantly improves fraud detection, intrusion detection, smart contract analysis, and anomaly detection.	Reviews supervised, unsupervised, reinforcement learning and deep learning techniques applied to blockchain.	Standardized datasets, reproducible benchmarks, explainable AI, and real-time blockchain deployment remain significant challenges.

[Practical No - 3]

Topic Given by Sir :

Formulation of Research Objectives and Hypotheses

Aim : To formulate the research problem, research questions, variables, research objectives, and hypotheses for studying the relationship between time spent watching short-form videos and students' ability to maintain attention during academic activities.

Q.1. Research Problem

The main research problem in this study is to find out whether the amount of time college students spend watching short-form videos has any relationship with their ability to concentrate during academic activities. Short-form video platforms such as Instagram Reels and YouTube Shorts have become a common source of entertainment among students, and students can easily spend a large amount of time watching one video after another.

At the same time, attention and concentration are important for students because activities such as attending long lectures, reading textbooks, understanding concepts, and completing assignments require students to remain focused for a certain period of time. If students are spending several hours every day watching short videos, it is important to understand whether this habit is associated with their ability to maintain attention during academic activities.

Therefore, the study will investigate the relationship between **daily time spent watching short-form videos** and **students' attention during academic activities**. The study will also compare students who spend different amounts of time watching short-form videos to determine whether their attention levels differ.

Q.2. Research Questions

Research Question 1

Is there a relationship between the amount of time college students spend watching shortform videos and their ability to maintain attention during academic activities?

Research Question 2

Do students who spend more time watching short-form videos have different attention levels compared with students who spend less time watching short-form videos?

Research Question 3

Does spending more than two hours per day watching short-form videos have any association with students' ability to concentrate during longer lectures or reading activities?

Q.3. Independent Variable

Time spent watching short-form videos per day.

The independent variable is the factor that is considered as a possible influence on the outcome of the study. In this case, the researcher is interested in the amount of time students spend watching platforms such as Instagram Reels and YouTube Shorts.

The time can be recorded in **minutes or hours per day**. For easier comparison, students can also be divided into groups based on their daily usage. For example, students may be classified as those who watch short-form videos for less than one hour, one to two hours, or more than two hours per day.

The researcher will then compare the attention levels of students with different amounts of short-form video usage.

Q.4. Dependent Variable

Students' ability to maintain attention during academic activities.

The dependent variable represents the outcome that the researcher wants to observe or measure. In this case, the researcher wants to understand how well students are able to maintain their concentration while performing academic activities.

Attention can be studied during activities such as:

- Attending a longer classroom lecture
- Reading academic material
- Listening to a teacher for an extended period
- Completing an academic task that requires concentration

The attention level may be represented using an attention or concentration score, depending on the method used by the researcher.

The dependent variable is expected to provide an indication of whether students with different short-form video viewing habits show differences in their ability to remain focused.

Q.5. Research Objectives

Objective 1

To determine the amount of time college students spend watching short-form videos such as Instagram Reels and YouTube Shorts on a daily basis.

Objective 2

To measure the ability of college students to maintain attention during academic activities such as longer lectures and reading activities.

Objective 3

To examine whether there is a relationship between the time spent watching short-form videos and students' ability to maintain attention during academic activities.

Objective 4

To compare the attention levels of students who spend different amounts of time watching short-form videos each day.

Q.6. Null Hypotheses (H_0)

Null Hypothesis 1 (H_{01})
There is no significant relationship between the amount of time college students spend watching short-form videos and their ability to maintain attention during academic activities.

Null Hypothesis 2 (H_{02})
There is no significant difference in the attention levels of students who spend different amounts of time watching short-form videos.

Q.7. Alternative Hypotheses (H_1)

Alternative Hypothesis 1 (H_{11})
There is a significant relationship between the amount of time college students spend watching short-form videos and their ability to maintain attention during academic activities.

Alternative Hypothesis 2 (H_{12})
There is a significant difference in the attention levels of students who spend different amounts of time watching short-form videos.

[Practical No - 3]

Topic Of My Research paper :

Research Formulation

Q.1 Research Problem Detection of Illicit Ethereum Accounts Using Explainable and FeatureEfficient Machine Learning

Ethereum is a widely used blockchain platform, but its transparent transaction history can also be used to study fraudulent and illicit activities such as scams, money laundering and other suspicious behavior. Traditional manual investigation of blockchain accounts is difficult because of the large amount of transaction data.

Previous research has successfully used Machine Learning, particularly XGBoost, to identify illicit Ethereum accounts. For example, Farrugia, Ellul and Azzopardi used **4,681 Ethereum accounts**, consisting of **2,179 illicit and 2,502 normal accounts**, and achieved an average accuracy of 96.3% using XGBoost with 10-fold cross-validation.

More recent research has improved the problem further. For example, the 2025 ETHIAD study combines **ADASYN, Lasso feature selection, XGBoost and SHAP**, specifically addressing class imbalance, feature selection and model interpretability.

Therefore, our research will investigate whether a machine-learning framework can provide **high detection performance while also reducing unnecessary features and explaining why an Ethereum account is classified as illicit**.

Q.2 Research Questions

Research Question 1
Which Machine Learning algorithm provides the best performance for detecting illicit Ethereum accounts?

Research Question 2
Can the number of Ethereum transaction features be reduced while maintaining a high level of illicit-account detection performance?

Research Question 3
Which transaction and ERC-20 behavioral features contribute most to XGBoost's prediction of illicit Ethereum accounts?

Q.3 Independent Variable

The independent variables are the factors that are changed or compared during the research. In this study, the main independent variables are the **Machine Learning algorithms, feature sets, and Ethereum transaction characteristics**.

The first independent variable is the **Machine Learning algorithm** used for classification.

Four algorithms are considered in the study: **XGBoost, Random Forest, Decision Tree and Support Vector Machine (SVM)**. Each algorithm works differently and may provide different results when applied to Ethereum transaction data. Comparing these algorithms helps determine which one provides better performance for illicit-account detection. The second independent variable is the **feature set**. The research compares the complete set of selected numeric features with a reduced feature set. The purpose of this comparison is to determine whether the number of features can be reduced without causing a large decrease in the model's detection performance.

The third independent variable consists of the **Ethereum account and transaction characteristics** used by the models. These include the number of transactions, amount of Ether received, amount of Ether sent, ERC-20 transaction count, number of unique receiving addresses, transaction timing and transaction values. These characteristics represent the behavior of an Ethereum account and are used by the Machine Learning algorithms to identify differences between normal and illicit accounts.

Q.4 Dependent Variable

The dependent variables are the results that are measured after applying the Machine Learning models. In this research, the main dependent variables are the **performance measures of the classification models**.

The first measure is **Accuracy**, which represents the percentage of accounts that are correctly classified as normal or illicit.

The second measure is **Precision**, which shows how many of the accounts predicted as illicit are actually illicit. Precision is useful because a detection system should avoid incorrectly identifying normal accounts as suspicious.

The third measure is **Recall**, which shows how many of the actual illicit accounts are successfully detected by the model. Recall is particularly important in security-related applications because failing to detect an illicit account can be more serious than incorrectly flagging a normal account.

The fourth measure is the **F1-score**, which combines Precision and Recall into a single measure. It is useful when both correctly identifying illicit accounts and avoiding false alarms are important.

The fifth measure is **ROC-AUC**, which measures how well the model can distinguish between normal and illicit accounts across different classification thresholds.

The research also uses **SHAP (SHapley Additive exPlanations)** for model interpretation. SHAP is used to identify which Ethereum transaction and account features have the greatest influence on the model's predictions. This makes the Machine Learning model more understandable because we can examine not only whether an account is classified as illicit, but also which features contributed to that prediction.

Q.5 Research Objectives Objective 1

To analyze Ethereum transaction and account-level behavioral features for identifying patterns associated with illicit accounts.

Objective 2

To implement and compare different Machine Learning algorithms for detecting illicit Ethereum accounts.

Objective 3

To investigate the effect of feature reduction on the performance and complexity of illicitaccount detection models.

Objective 4

To evaluate the selected models using Accuracy, Precision, Recall, F1-score and ROC-AUC through stratified cross-validation.

Objective 5

To use SHAP-based Explainable AI to identify and interpret the most influential features involved in illicit-account detection.

Q.6 Null Hypotheses (H_0)

Null Hypothesis 1 (H_{01})

There is no significant difference in the detection performance of the selected Machine Learning algorithms for identifying illicit Ethereum accounts.

Null Hypothesis 2 (H_{02})

Reducing the number of transaction-related features does not have a significant effect on the performance of Machine Learning models used for detecting illicit Ethereum accounts.

Q.7 Alternative Hypotheses (H_1)

Alternative Hypothesis 1 (H_{11})

There is a significant difference in the detection performance of the selected Machine Learning algorithms for identifying illicit Ethereum accounts.

Alternative Hypothesis 2 (H_{12})

Reducing the number of transaction-related features has a significant effect on the performance of Machine Learning models used for detecting illicit Ethereum accounts.

[Practical No - 4]
Explainable and Feature-Efficient Detection of Illicit Ethereum Accounts Using Machine Learning

Paper 1 :

Detection of illicit accounts over the Ethereum blockchain. Farrugia, S., Ellul, J., & Azzopardi, G. (2020). *Expert Systems with Applications, 150*, 113318.

1. Research Problem / Topic Addressed

The paper addresses the problem of detecting illicit accounts on the Ethereum blockchain. Since blockchain transaction information is publicly available, it can be analyzed to identify accounts associated with suspicious or illicit activities. However, manually analyzing thousands of accounts and their transaction histories is difficult. Therefore, the study investigates whether Machine Learning can automatically distinguish illicit accounts from normal accounts.

2. Objectives of the Study

The main objective is to develop a Machine Learning-based approach for identifying illicit Ethereum accounts. The study also investigates which transaction-history features are useful for distinguishing illicit accounts from normal accounts.

3. Research Methodology Used

The researchers extracted transaction-history features from Ethereum accounts and used supervised Machine Learning algorithms to classify accounts into illicit and normal categories. Different models were evaluated using classification performance measures.

4. Dataset / Sample Used

The dataset contains **4,681 Ethereum accounts**, consisting of **2,179 illicit accounts and 2,502 normal accounts**.

5. Tools / Techniques / Algorithms Used

- Machine Learning
- XGBoost
- Transaction-history features
- Feature analysis
- Classification

6. Major Findings

The study demonstrated that Machine Learning can successfully identify illicit Ethereum accounts. XGBoost achieved strong classification performance and showed that transactionhistory characteristics contain useful information for identifying suspicious accounts.

7. Limitations / Research Gaps

The study primarily focuses on classification performance. Further research can investigate feature efficiency, explainability and whether a smaller set of important features can maintain similar detection performance.

8. Relevance to Our Proposed Research

This paper forms the main foundation of our research because our work also focuses on detecting illicit Ethereum accounts using Machine Learning. However, our research extends this direction by comparing multiple Machine Learning algorithms and investigating the performance trade-off between full and reduced feature sets.

9. Important Learning from the Paper

The important learning is that Ethereum transaction-history features can be used effectively for identifying illicit accounts using Machine Learning.

Paper 2

Detecting anomalies in blockchain transactions using machine learning classifiers and explainability analysis.

Hasan, M., Rahman, M. S., Janicke, H., & Sarker, I. H. (2024). *Blockchain: Research and Applications*, *5*, 100207.

1. Research Problem / Topic Addressed

The paper focuses on detecting anomalous activities in blockchain transactions. The large volume of blockchain transactions makes manual detection difficult, creating a need for automated Machine Learning-based detection systems.

2. Objectives of the Study

The study aims to evaluate Machine Learning classifiers for detecting anomalous blockchain transactions and investigate the contribution of different features using Explainable AI.

3. Research Methodology Used

The researchers preprocess blockchain transaction data and apply multiple Machine Learning classification techniques. The models are evaluated using standard performance metrics. Explainability analysis is then used to identify important features influencing model predictions.

4. Dataset / Sample Used

The study uses blockchain transaction data for anomaly detection.

5. Tools / Techniques / Algorithms Used

- Random Forest
- XGBoost
- Ensemble models
- SHAP
- Feature analysis
- Classification

6. Major Findings

The research demonstrates that Machine Learning and ensemble approaches can effectively identify anomalous blockchain transactions. Explainability analysis provides additional information about the factors influencing model decisions.

7. Limitations / Research Gaps

Further research is required for generalized datasets, real-time transaction detection and evaluation across different blockchain environments.

8. Relevance to Our Proposed Research

This paper strongly supports our use of both **Machine Learning and Explainable AI**. However, our research focuses specifically on Ethereum illicit accounts and additionally studies whether feature reduction can maintain high model performance.

9. Important Learning from the Paper

A blockchain security model should not only provide accurate predictions but should also provide an explanation of the factors responsible for those predictions.

Paper 3 :

Detecting anomalies in blockchain transactions using machine learning classifiers and explainability analysis.

Kang, Y., Kim, W., Kim, H., Lee, M., Song, M., & Seo, H. (2023). *Electronics*, *12*(18), 3893.

1. Research Problem / Topic Addressed

The paper focuses on identifying malicious smart contracts on blockchain networks. Malicious contracts can cause security and financial risks to blockchain users.

2. Objectives of the Study

The study aims to develop an efficient deep-learning model for detecting malicious smart contracts and reduce the computational requirements of the detection model.

3. Research Methodology Used

Smart-contract bytecode is converted into opcode information and opcode frequencies are used as input features. A deep-learning model is trained for classification, followed by Explainable AI analysis to identify important features.

4. Dataset / Sample Used

The study uses smart-contract opcode information representing benign and malicious contracts.

5. Tools / Techniques / Algorithms Used

- Deep Learning
- Opcode analysis
- Integrated Gradients
- Gradient SHAP
- Explainable AI
- Lightweight model

6. Major Findings

The proposed lightweight model achieved high detection performance while reducing the size and complexity of the model.

7. Limitations / Research Gaps

The approach focuses on malicious smart-contract detection rather than illicit account detection. Further research is required for broader blockchain security problems and realworld deployment.

8. Relevance to Our Proposed Research

The paper is relevant because it demonstrates the value of **feature reduction and Explainable AI** in blockchain security. These concepts are directly related to our featureefficient and explainable Ethereum-account detection framework.

9. Important Learning from the Paper

Reducing unnecessary information can improve the efficiency of a security model while maintaining useful detection performance.

Paper 4 :

A survey on the security of blockchain systems.

Li, X., Jiang, P., Chen, T., Luo, X., & Wen, Q. (2018).

1. Research Problem / Topic Addressed

The paper provides a comprehensive overview of security threats affecting blockchain systems.

2. Objectives of the Study

The objective is to identify major blockchain security vulnerabilities and review existing protection mechanisms.

3. Research Methodology Used

The study uses a survey-based methodology to classify blockchain attacks and security mechanisms.

4. Dataset / Sample Used

No experimental dataset is used because this is a survey paper. **5. Tools / Techniques / Algorithms Used** The study discusses:

- 51% attacks
- Selfish mining
- Double spending
- Eclipse attacks
- Smart-contract attacks
- Consensus attacks

6. Major Findings

The paper shows that blockchain security involves several layers and that different types of attacks require different protection mechanisms.

7. Limitations / Research Gaps

The study identifies continuing challenges in blockchain scalability, privacy, consensus security and smart-contract security.

8. Relevance to Our Proposed Research

This paper provides the theoretical foundation for our blockchain-security research and helps explain why automated detection of suspicious blockchain activity is required.

9. Important Learning from the Paper

Blockchain is secure by design but is not completely immune to attacks and malicious behavior.

Paper 5 :

Blockchain data security: A survey.

Zhu, L., Zheng, B., Shen, M., Yu, S., Gao, F., Li, H., Shi, K., & Gai, K. (2018).

1. Research Problem / Topic Addressed

The paper investigates different security problems associated with blockchain data, including privacy, integrity and availability.

2. Objectives of the Study

The objective is to classify blockchain data-security problems and examine possible solutions.

3. Research Methodology Used

The study reviews existing research related to blockchain data security and categorizes different security threats.

4. Dataset / Sample Used

No experimental dataset is used because the paper is a survey.

5. Tools / Techniques / Algorithms Used

The paper discusses:

- Privacy protection
- Data integrity
- Availability
- Secure data sharing
- Smart-contract security

6. Major Findings

The study identifies multiple challenges related to maintaining security and privacy in blockchain-based systems.

7. Limitations / Research Gaps

Privacy, secure data management and smart-contract security remain important research areas.

8. Relevance to Our Proposed Research

The paper establishes the broader blockchain-security background for our research. Our work addresses a specific security problem by detecting potentially illicit Ethereum accounts.

9. Important Learning from the Paper

Blockchain security includes both protection of stored information and detection of malicious activities.

Paper 6 :

AI + Blockchain Security

1. Research Problem / Topic Addressed

The paper examines how Artificial Intelligence can be used together with blockchain technology to improve automated security and threat detection.

2. Objectives of the Study

The objective is to study the application of AI techniques in blockchain security, including fraud detection, anomaly detection and smart-contract analysis.

3. Research Methodology Used

The study reviews existing research involving AI and blockchain-security applications.

4. Dataset / Sample Used

No single experimental dataset is used because the study is a review.

5. Tools / Techniques / Algorithms Used

- Machine Learning
- Deep Learning
- Anomaly Detection
- Fraud Detection
- Explainable AI
- Smart-contract analysis

6. Major Findings

AI can improve automated detection of suspicious activities and security threats in blockchain environments.

7. Limitations / Research Gaps

Important gaps include explainability, standardized datasets, efficient AI models and realworld validation.

8. Relevance to Our Proposed Research

The paper supports our decision to combine Machine Learning with Explainable AI for Ethereum security.

9. Important Learning from the Paper

Future blockchain-security systems need to be accurate, explainable and computationally efficient.

Paper 7 :
Graph Neural Network for Ethereum Fraud Detection

1. Research Problem / Topic Addressed
The research investigates Ethereum fraud detection using graph-based Machine Learning. Traditional models may not fully capture relationships between blockchain accounts.

2. Objectives of the Study
The objective is to use relationships between Ethereum accounts and transactions to improve fraud detection.

3. Research Methodology Used
Ethereum transactions are represented as a graph and Graph Neural Network techniques are applied to learn patterns associated with fraudulent behavior.

4. Dataset / Sample Used
Ethereum account and transaction relationship data.

5. Tools / Techniques / Algorithms Used
- Graph Neural Networks
- Graph representation
- Temporal transaction information
- Fraud detection

6. Major Findings
Graph-based approaches can capture relationships between blockchain accounts that may not be represented by traditional tabular features.

7. Limitations / Research Gaps
Graph-based models can require significant computational resources and may face scalability challenges with large blockchain datasets.

8. Relevance to Our Proposed Research
This paper provides an alternative approach to our feature-based Machine Learning framework. It also gives us a possible direction for future extension of our research.

9. Important Learning from the Paper
Blockchain accounts should be considered not only individually but also in terms of their relationships with other accounts.

Paper 8 :
Blockchain Attack Classification

1. Research Problem / Topic Addressed
The study focuses on identifying and classifying different types of attacks that can occur at different blockchain layers.

2. Objectives of the Study
The objective is to provide a systematic classification of blockchain attacks and their corresponding security mechanisms.

3. Research Methodology Used
The researchers categorize blockchain attacks according to different architectural layers.

4. Dataset / Sample Used
The study is primarily a classification and review-based study.

5. Tools / Techniques / Algorithms Used
The study considers attacks involving:

- Network layer
- Consensus layer
- Data layer
- Smart contracts
- Applications

6. Major Findings
The research demonstrates that blockchain attacks can occur at multiple layers and require different detection and mitigation strategies.

7. Limitations / Research Gaps
Automated, real-time and AI-based multi-layer attack detection remains an area requiring further research.

8. Relevance to Our Proposed Research
This paper helps define the broader blockchain-security landscape and allows us to position illicit Ethereum-account detection as a specific security problem.

9. Important Learning from the Paper
A blockchain-security research project should clearly identify the specific security threat being studied.

Paper 9 :
Blockchain + Machine Learning Survey

1. Research Problem / Topic Addressed
The paper reviews the application of Machine Learning and Artificial Intelligence techniques to different blockchain problems.

2. Objectives of the Study
The objective is to investigate how Machine Learning can be used for blockchain applications including fraud detection, anomaly detection and cybersecurity.

3. Research Methodology Used
The research reviews existing Machine Learning and blockchain studies.

4. Dataset / Sample Used
No single experimental dataset is used.

5. Tools / Techniques / Algorithms Used
The study covers:

- Supervised Learning
- Unsupervised Learning
- Deep Learning
- Anomaly Detection
- Fraud Detection
- Smart-contract analysis

6. Major Findings
Machine Learning provides useful approaches for detecting suspicious blockchain behavior and improving blockchain security.

7. Limitations / Research Gaps
The literature indicates a need for standardized datasets, reproducible experiments, explainability and real-time evaluation.

8. Relevance to Our Proposed Research
This paper supports our Machine Learning approach and reinforces the importance of comparing models using standardized evaluation metrics.

9. Important Learning from the Paper
Machine Learning is promising for blockchain security, but proper evaluation and explainability are important for reliable results.

Paper 10 :

Blockchain and Explainable AI for Enhanced Decision Making in Cyber Threat Detection

1. Research Problem / Topic Addressed

The study addresses the problem of limited transparency in AI-based cybersecurity systems. A Machine Learning model may correctly identify a threat but may not explain why the threat was detected.

2. Objectives of the Study

The objective is to investigate how Explainable AI can improve transparency and decisionmaking in cyber threat detection.

3. Research Methodology Used

The research combines blockchain-based security concepts with Explainable AI techniques to improve the transparency of security decisions.

4. Dataset / Sample Used

The study focuses on cybersecurity detection rather than specifically using our Ethereum illicit-account dataset.

5. Tools / Techniques / Algorithms Used

- Blockchain
- Explainable AI
- Cyber threat detection
- AI-based security analysis

6. Major Findings

Explainable AI can improve the transparency and interpretability of AI-based security systems.

7. Limitations / Research Gaps

Further research is required for lightweight XAI systems and evaluation using real blockchain transaction datasets.

8. Relevance to Our Proposed Research

This paper directly supports our use of **SHAP**. Our research will not only classify Ethereum accounts but will also investigate which transaction features influence the classification.

9. Important Learning from the Paper

For blockchain security, a model should ideally explain its predictions instead of functioning only as a black box.

[Practical No - 5]

“Impact of Social Media Usage on Academic Performance of College Students.”

1. Research Problem Statement :

Social media has become an important part of students' daily lives. College students use platforms such as Instagram, YouTube, Facebook, Snapchat and other social networking applications for communication, entertainment, information sharing and educational purposes. However, spending excessive time on social media may reduce the amount of time students devote to studying and completing academic activities. It may also affect their concentration, time management and ability to maintain attention during academic work.

Therefore, this research aims to study whether the amount of time college students spend using social media has an impact on their academic performance. The research will examine students' social media usage patterns and compare them with their academic performance to determine whether a significant relationship exists between the two variables.

2. Research Objectives :

The main objectives of this research are:

1. To study the social media usage patterns of college students.
2. To determine the average amount of time students spend using social media each day.
3. To examine the academic performance of college students.
4. To investigate whether social media usage has a significant relationship with students' academic performance.
5. To compare the academic performance of students with different levels of social media usage.
6. To determine whether excessive social media usage is associated with lower academic performance.

3. Research Questions / Hypotheses :

Research Question 1
How much time do college students spend using social media on a daily basis?
Research Question 2
What is the academic performance level of college students?
Research Question 3
Is there a significant relationship between social media usage and academic performance?
Research Question 4
Is there a significant difference in academic performance among students with different levels of social media usage?

Null Hypotheses (H_0) :

H_{01}: There is no significant relationship between the amount of time college students spend using social media and their academic performance.

H_{02}: There is no significant difference in the academic performance of students who spend different amounts of time using social media.

Alternative Hypotheses (H_1) :

H_{11}: There is a significant relationship between the amount of time college students spend using social media and their academic performance.

H_{12}: There is a significant difference in the academic performance of students who spend different amounts of time using social media.

4. Research Approach :

The research will follow a **quantitative research approach** because numerical data will be collected from college students regarding their social media usage and academic performance.

The collected data will be statistically analyzed to determine the relationship between social media usage and academic performance.

5. Research Design :

The study will use a **descriptive and correlational research design**.

The descriptive design will be used to understand students' social media usage patterns and academic performance.

The correlational design will be used to determine whether there is a relationship between the amount of time students spend on social media and their academic performance.

The study will not manipulate students' social media usage. Instead, information about their existing usage and academic performance will be collected and analyzed.

6. Identify the Population and Decide the Sample :

Population

The population of the study will consist of **college students**.

Sample

A sample of approximately **100 college students** will be selected for the study.

Students from different academic years and backgrounds may be included so that the sample represents different levels of social media usage and academic performance.

7. Select Sampling Technique :

The study will use **convenience sampling** because college students who are readily available and willing to participate will be selected for the research.

The participants will be informed about the purpose of the research before responding to the questionnaire

8. Identify the Variables :

Independent Variable :

Social Media Usage

This will be measured using factors such as:

- Daily time spent on social media
- Frequency of social media usage
- Number of social media platforms used
- Purpose of using social media

The primary independent variable will be **daily time spent on social media**.

Dependent Variable :

Academic Performance

Academic performance can be measured using:

- Percentage/marks
- CGPA
- GPA
- Recent semester performance

The student's academic score will be used as the main indicator of academic performance.

9. Decide the Data Collection Method :

The research will use **primary data collection**.

Data will be collected directly from college students using a structured questionnaire.

The questionnaire will collect information about:

- Students' social media usage
- Daily time spent on social media
- Frequency of usage
- Purpose of usage
- Academic performance

The questionnaire can be distributed electronically using an online form.

10. Design the Data Collection Instrument :

The main data collection instrument will be a **structured questionnaire**.

The questionnaire will contain questions divided into different sections.

Section A – Basic Information

- Age
- Gender
- Academic year
- Course/program

Section B – Social Media Usage

- Which social media platforms do you use?
- How many hours do you spend on social media each day?
- How frequently do you check social media?
- What is your main purpose for using social media?
- Do you use social media while studying?

Section C – Academic Performance

- Current/latest CGPA or percentage
- Average study hours per day
- Frequency of completing academic work
- Self-assessment of academic performance

A Likert-scale format can also be used for questions related to concentration, study habits and academic activities.

11. Decide the Data Analysis Technique :

The collected data will be analyzed using statistical techniques.

Descriptive Statistics :

Descriptive statistics will be used to summarize the collected data.

The following measures can be calculated:

- Mean
- Median
- Standard deviation
- Frequency
- Percentage

Correlation Analysis :

Pearson Correlation can be used to determine the relationship between daily social media usage and academic performance when the relevant assumptions are satisfied.

The correlation coefficient will indicate whether the relationship is positive, negative or weak/strong.

Group Comparison :

Students can be divided into groups according to their daily social media usage, for example:

- Low usage
- Moderate usage
- High usage

Their academic performance can then be compared using an appropriate statistical test such as **ANOVA**, depending on the data and assumptions.

Significance Level

A significance level of **0.05 (5%)** can be used to determine whether the observed relationship or difference is statistically significant.

12. Select the Tools / Software :

Google Forms : Collect questionnaire responses

Microsoft Excel : Store and organize collected data

Python : Statistical analysis and visualization

Pandas : Data processing

NumPy : Numerical calculations

Matplotlib : Graphs and visualization

Seaborn : Statistical visualization

SciPy : Statistical tests

13. Ethical Considerations :

The research will follow basic ethical principles while collecting information from college students.

Participation will be **voluntary**, and participants will be informed about the purpose of the research. No participant will be forced to provide information.

Personal information that is not required for the research will not be collected. The collected responses will be kept confidential and will be used only for academic research purposes.

Participants should also be informed that they can choose not to participate or withdraw their response where applicable.

The research results will be reported honestly without changing or manipulating the collected data.

14. Proposed Research Procedure :

The complete research procedure will be carried out as follows:

Step 1 – Identify Research Problem

Identify the problem of excessive or varying social media usage and its possible relationship with students' academic performance.

Step 2 – Prepare Research Objectives

Define what the study intends to investigate.

Step 3 – Prepare Questionnaire

Design questions related to social media usage and academic performance.

Step 4 – Select Participants

Select approximately 100 college students using convenience sampling.

Step 5 – Collect Data

Distribute the questionnaire and collect responses.

Step 6 – Organize Data

Export the responses into Excel/Python and clean the collected data.

Step 7 – Descriptive Analysis

Calculate frequencies, percentages, mean and standard deviation.

Step 8 – Correlation Analysis

Analyze the relationship between social media usage and academic performance.

Step 9 – Group Comparison

Compare academic performance among students with different levels of social media usage.

Step 10 – Hypothesis Testing

Test the null and alternative hypotheses using the appropriate statistical methods.

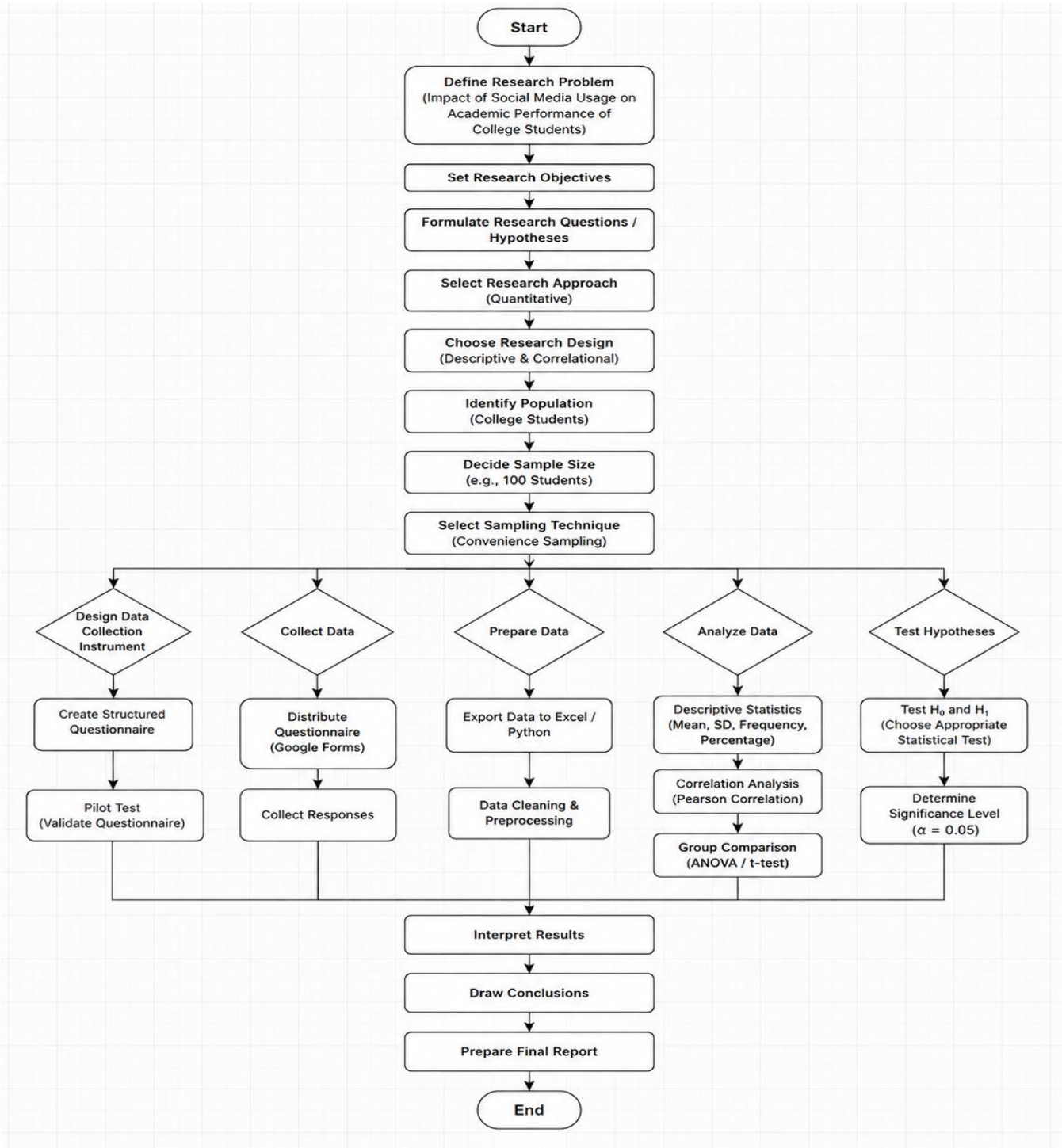
Step 11 – Interpret Results

Determine whether the findings indicate a significant relationship or difference.

Step 12 – Conclusion

Draw a conclusion about the impact/relationship between social media usage and academic performance.

15. Complete Flow Chart :



[Practical No - 6]

Pilot Data Collection and Software Framework Design

Research Topic : Explainable and Feature-Efficient Machine Learning for Illicit Ethereum Account Detection

Aim : To design a pilot data collection procedure and software framework for detecting illicit Ethereum accounts using machine learning, while studying feature efficiency and explaining model predictions using SHAP.

Research Problem

Illicit activity can occur through Ethereum accounts, and manual identification of suspicious account behavior is difficult because transaction histories contain many behavioral features. This research proposes to evaluate Machine Learning models for classifying Ethereum accounts as normal or illicit, while investigating whether a reduced set of features can retain useful detection performance and whether SHAP can make the model's decisions more interpretable.

Note on the data source

The benchmark used for this practical is the dataset associated with Farrugia, Ellul and Azzopardi's 2020 study. The authors describe 2,179 illicit accounts and 2,502 normal accounts, giving 4,681 accounts, with 42 extracted features per account. The dataset was published as a benchmark for related work.

Source: Farrugia, S., Ellul, J., & Azzopardi, G. (2020), *Detection of illicit accounts over the Ethereum blockchain*, Expert Systems with Applications, 150, 113318. DOI:

10.1016/j.eswa.2020.113318.

TASK A – PILOT DATA COLLECTION

1. Research Question

Can account-level transaction and behavioural features be used to effectively distinguish illicit Ethereum accounts from normal Ethereum accounts, and can a reduced feature set maintain useful classification performance while providing interpretable predictions?

2. Population, Sampling Technique and Proposed Sample Size

Item	Details
Population	Ethereum accounts represented in the selected labeled benchmark dataset.
Sampling technique	Secondary-data selection from the published benchmark dataset. For the pilot, a small anonymised subset is used to check the data
Proposed full sample	4,681 labeled account records: 2,179 illicit and 2,502 normal accounts.
Pilot sample	15 records: 8 illicit and 7 normal accounts, selected from the benchmark CSV for pilot data checking only.

3. Data Collection Instrument

Because this is a secondary-data Machine Learning study, a questionnaire for human participants is not appropriate. The practical's data-collection instrument is therefore implemented as a structured **Ethereum Account Data Extraction Sheet**. The sheet records the fields required for model preparation and the target account class.

No.	Field / check	Purpose
1	Account class	Normal or illicit target label.
2	Transaction counts	Sent, received and total transaction activity.
3	Transaction timing	Average intervals and account active duration.
4	Ether values	Minimum, maximum, average, sent, received and balance values.

5	Contract activity	Contract creation and Ether sent to contracts.
6	ERC20 activity	Token transaction counts, values and counterparties.
7	Data completeness	Check blank, missing and invalid values.
8	Duplicate check	Check repeated account records.
9	Feature usability	Identify identifier/non-predictive fields before modeling.
10	Pilot notes	Record issues discovered during the pilot.

4. Pilot Sample Dataset

The following 15 records are a pilot subset taken from the published benchmark CSV. The original Ethereum addresses are not reproduced here; they are replaced by Pilot IDs. The pilot is used to demonstrate data entry and quality checking and must not be treated as a population-level result.

Pilot_ID	Account_Class	Avg sent (min)	Avg received (min)	Active time (min)	Sent Tnx	Received Tnx	Total Tnx	Ether sent	Ether received	Ether balance	ERC20 Tnx
P-01	Illicit	1457.31	34.12	4815.43	3	13	16	17.52874745	17.52978485	0.001037 4	
P-02	Illicit	3976.5	834.77	9622.53	2	2	4	1.53269261	1.53378461	0.001092	1.0
P-03	Illicit	112.9	31.87	321.42	2	3	5	1.317896	1.31882	0.000924	1.0
P-04	Illicit	2300.37	65.1	73091.0	29	98	127	38.13376706	37.55604606	-0.5777 21002	96.0
P-05	Illicit	0.0	0.0	0.0	0	0	0	0.0	0.0	0.0	
P-06	Illicit	59129.36	0.0	118258.72	2	1	3	0.6793	0.68	0.0007	1.0
P-07	Illicit	1819.9	638.12	8744.77	2	8	10	2.74925268	2.75097468	0.001722	1.0
P-08	Illicit	6556.08	255.89	53728.05	8	5	13	116.996572	117.0001	0.003528	1.0
P-09	Normal	844.26	1093.71	704785.63	721	89	810	865.6910932	586.4666748	-279.22 44185	265.0
P-10	Normal	12709.07	2958.44	1218216.73	94	8	102	3.08729702	3.085478209	-0.0018 18811	8.0

P-11	Normal	163.78	1.11	329.78	2	2	4	100.9989738	101.0	0.0010 26218	0.0
P-12	Normal	1067.84	128.81	636162.82	68	4375	4443	4618.790832	4618.823828	0.0329 9662	10.0
P-13	Normal	2.12	0.92	21206.93	9996	4	10003	0.0	44.0	44.0	34.0
P-14	Normal	164.22	0.0	328.45	2	1	3	100.9989738	101.0	0.0010 26218	0.0
P-15	Normal	20827.93	0.0	186707.1	8	1	9	101.5119377	10.0	- 91.511 93767	1.0

5. Pilot Data Quality Check

Check	Pilot observation	Action for full study
Missing values	Blank values are present in some ERC20-related fields in the	raw CSV.
Duplicate entries	The pilot rows have distinct record indices.	Check the full dataset for duplicate account record before model training.
Unclear answers	Not applicable; the dataset is structured numerical/categorical	datValidate column types and categorical token fields..
Identifier fields	Account addresses identify records but should not be used as	ordinary predictive Keep identifiers only for traceability; exclude them from model features.ume ic features.
Class field	The pilot contains both normal and illicit records.	Preserve the target label separately from input features.

6. What the Pilot Revealed

The pilot confirms that the benchmark can be represented as account-level structured records with a binary class label and multiple transaction-behavior features. It also shows that ERC20-related fields can contain blank values, so the preprocessing stage should explicitly define how missing values are handled. The pilot does not establish model accuracy or population-level behavior; those questions belong to the full experimental analysis.

TASK B — SOFTWARE FRAMEWORK DESIGN 1. Research Problem

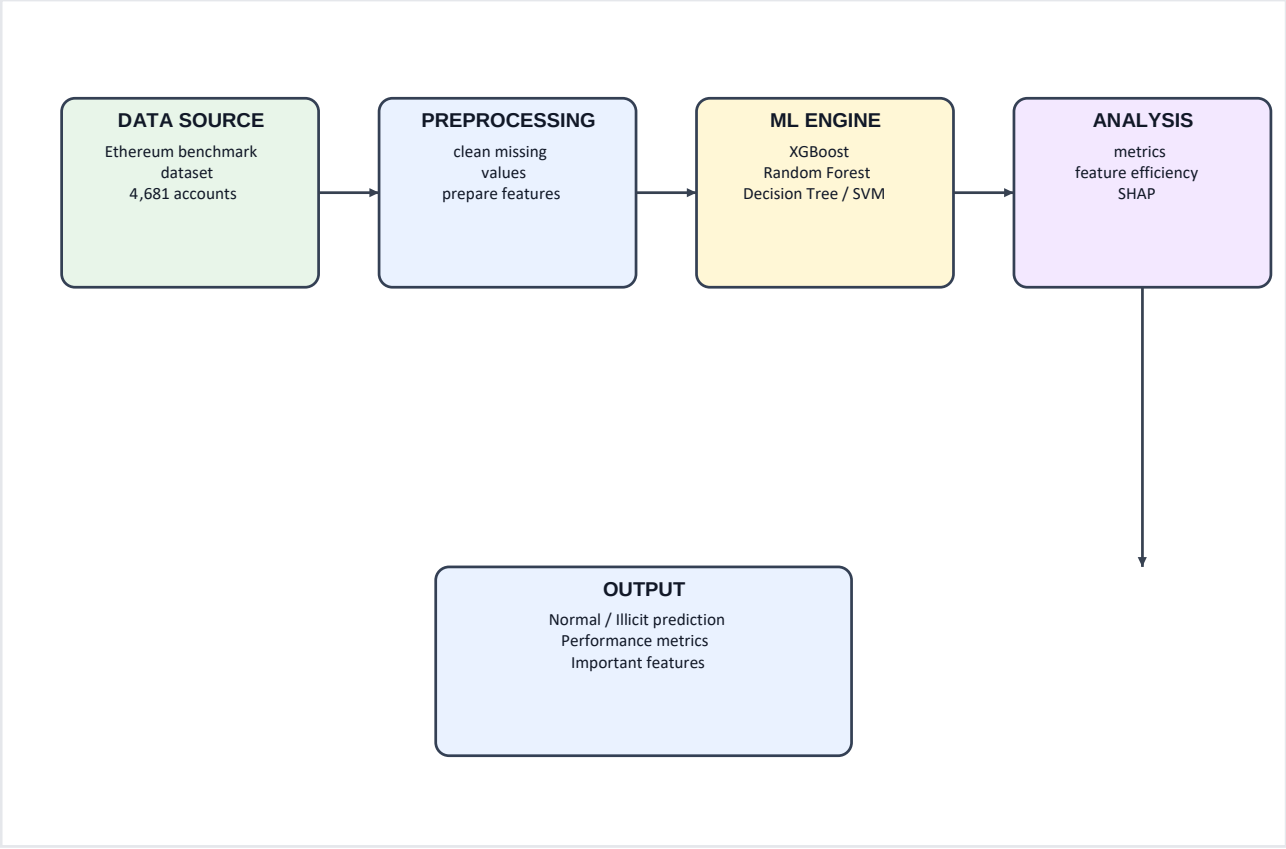
1. Research Problem

Design a Machine Learning framework that classifies Ethereum accounts as normal or illicit using transaction-history features, compares full and reduced feature sets, evaluates several classifiers, and provides feature-level explanations for the selected model using SHAP.

2. Functional Requirements

FR1	Data Input	Accept the labeled Ethereum account dataset and required feature fields.
FR2	Preprocessing	Validate, clean and prepare the input features for Machine Learning.
FR3	Classification	Train and evaluate XGBoost, Random Forest, Decision Tree and SVM models.
FR4	Feature Efficiency	Compare full and reduced feature sets using the same evaluation procedure.
FR5	Explainability	Use SHAP on the selected model to identify influential features and explain predictions.

3. Input → Processing → Output Block Diagram



4. Main Modules and Their Purpose

Module	Purpose
1. Data Input Module	Loads the Ethereum account dataset and identifies the target class and candidate features.
2. Data Preprocessing Module	Checks data types, missing values, duplicates and unsuitable identifier fields; prepares the modeling matrix.
3. Feature Preparation Module	Creates the full feature set and the reduced feature set used for comparison.
4. Machine Learning Module	Trains XGBoost, Random Forest, Decision Tree and SVM classifiers.
5. Model Evaluation Module	Uses Stratified 5-Fold Cross-Validation and calculates Accuracy, Precision, Recall, F1-score and ROC-AUC.
6. Explainable AI Module	Applies SHAP to the selected model to show which features influence predictions.
7. Reporting Module	Presents model comparison, feature-efficiency findings and explainability results.

5. Data Sources, Inputs, Outputs and Technology Stack

Component	Proposed detail
Data source	Published Ethereum illicit-account benchmark dataset; original study combined EtherscamDB labels with Ethereum transaction data and
Input fields	42 extracted account-level transaction/behavior features plus the target class; account identifier retained separately for traceability.
Processing	Data validation → missing-value handling → feature preparation → full/reduced feature comparison → model training.
Expected output	Normal/illicit class prediction, model metrics, full-vs-reduced feature comparison, SHAP feature contributions.
Technology	Python, Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Matplotlib/Seaborn, Google Colab or Jupyter Notebook.

6. Important Workflow / Interface Sketch

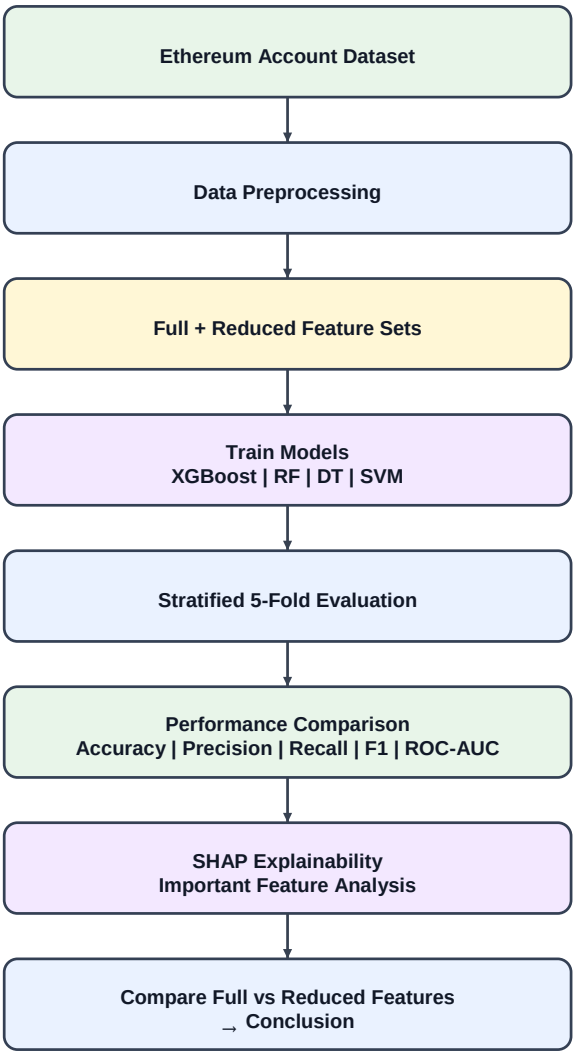
User/System Action	Framework Response
Upload / select Ethereum account data	Validate file structure and identify required columns.
Run preprocessing	Clean data and prepare model-ready features.
Select feature configuration	Run full-feature or reduced-feature pipeline.
Run detection	Generate model prediction for normal/illicit class.
View evaluation	Display Accuracy, Precision, Recall, F1-score and ROC-AUC.
View explanation	Display SHAP feature contribution for the selected model/prediction.

ETHEREUM ILLICIT ACCOUNT DETECTION — PROPOSED WORKFLOW
[Upload Dataset] [Select Feature Set: Full / Reduced] [Run Analysis]
Prediction: NORMAL / ILLICIT Metrics: Accuracy | Precision | Recall | F1 | ROC-AUC
Explanation: SHAP Feature Importance / Individual Prediction Explanation

7. Framework Evaluation

Evaluation area	Measure / method	Purpose
Classification performance	Accuracy, Precision, Recall, F1-score, ROC-AUC	Measure how well the models distinguish normal and illicit accounts.
Cross-validation stability	Stratified 5-Fold CrossValidation	Reduce dependence on one train/test split and preserve class proportions.
Feature efficiency	Compare full vs reduced feature set	Determine whether fewer features can maintain comparable performance.
Explainability	SHAP global and local explanations	Identify influential features and explain individual predictions.
Computational efficiency	Training/inference time and feature count	Document whether feature reduction changes computational requirements.

8. Proposed Research Procedure



APPENDIX A — DATA DICTIONARY

The following dictionary follows the 42 extracted features reported in the benchmark paper's Appendix Table A1. The paper describes these as account-level features extracted from Ethereum transactions.

#	Variable	Meaning	Type
1	Avg_min_between_sent_tnx	Average time between sent transactions for the account (minutes)	Integer
2	Avg_min_between_received_tnx	Average time between received transactions for the account (minutes)	Integer
3	Time_Diff_between_first_and_last(Mins)	Time difference between the first and last transaction	Integer
4	Sent_tnx	Total number of sent normal transactions	Integer
5	Received_tnx	Total number of received normal transactions	Integer
6	Number_of_Created_Contracts	Total number of created contract transactions	Integer
7	Unique_Received_From_Addresses	Total unique addresses from which the account received transactions	Integer
8	Unique_Sent_To_Addresses	Total unique addresses to which the account sent transactions	Integer
9	Min_Value_Received	Minimum value in Ether ever received	Double
10	Max_Value_Received	Maximum value in Ether ever received	Double
11	Avg_Value_Received	Average value in Ether ever received	Double
12	Min_Val_Sent	Minimum value of Ether ever sent	Double
13	Max_Val_Sent	Maximum value of Ether ever sent	Double
14	Avg_Val_Sent	Average value of Ether ever sent	Double
15	Min_Value_Sent_To_Contract	Minimum value of Ether sent to a contract	Double
16	Max_Value_Sent_To_Contract	Maximum value of Ether sent to a contract	Double
17	Avg_Value_Sent_To_Contract	Average value of Ether sent to contracts	Double
18	Total_Transactions(Including_Tnx_to_Create_Contract)	Total number of transactions	Integer
19	Total_Ether_Sent	Total Ether sent for the account address	Double
20	Total_Ether_Received	Total Ether received for the account address	Double

21	Total_Ether_Sent_Contracts	Total Ether sent to contract addresses	Double
22	Total_Ether_Balance	Total Ether balance following enacted transactions	Double
23	Total_ERC20_Tnxs	Total number of ERC20 token transfer transactions	Integer
24	ERC20_Total_Ether_Received	Total ERC20 token received transactions in Ether	Double
25	ERC20_Total_Ether_Sent	Total ERC20 token sent transactions in Ether	Double
26	ERC20_Total_Ether_Sent_Contract	Total ERC20 token transfer to other contracts in Ether	Double
27	ERC20_Uniq_Sent_Addr	Number of ERC20 token transactions sent to unique account addresses	Integer
28	ERC20_Uniq_Rec_Addr	Number of ERC20 token transactions received from unique addresses	Integer
29	ERC20_Uniq_Rec_Contract_Addr	Number of ERC20 token transactions received from unique contract addresses	Integer
30	ERC20_Avg_Time_Between_Sent_Tnx	Average time between ERC20 token sent transactions (minutes)	Integer
31	ERC20_Avg_Time_Between_Rec_Tnx	Average time between ERC20 token received transactions (minutes)	Integer
32	ERC20_Avg_Time_Between_Contract_Tnx	Average time between ERC20 token sent/contract transactions (minutes)	Integer
33	ERC20_Min_Val_Rec	Minimum value in Ether received from ERC20 token transactions	Double
34	ERC20_Max_Val_Rec	Maximum value in Ether received from ERC20 token transactions	Double
35	ERC20_Avg_Val_Rec	Average value in Ether received from ERC20 token transactions	Double
36	ERC20_Min_Val_Sent	Minimum value in Ether sent from ERC20 token transactions	Double
37	ERC20_Max_Val_Sent	Maximum value in Ether sent from ERC20 token transactions	Double

38	ERC20_Avg_Val_Sent	Average value in Ether sent from ERC20 token transactions	Double
39	ERC20_Uniq_Sent_Token_Name	Number of unique ERC20 tokens transferred	Integer
40	ERC20_Uniq_Rec_Token_Name	Number of unique ERC20 tokens received	Integer
41	ERC20_Most_Sent_Token_Type	Most sent token for the account via ERC20 transaction	String
42	ERC20_Most_Rec_Token_Type	Most received token for the account via ERC20 transaction	String

OBSERVATIONS

The pilot confirms that the selected Ethereum benchmark is suitable for a structured Machine Learning workflow because it contains labeled normal and illicit accounts together with transaction-derived account features. The pilot also demonstrates why data validation is necessary before model training: some ERC20-related fields may be blank, while identifier and target fields must be treated differently from predictive features. The proposed framework therefore separates data validation, feature preparation, model comparison and explainability rather than treating the raw CSV as immediately model-ready.

CONCLUSION

Practical 06 establishes a pilot data-collection and software-framework plan for the research problem of illicit Ethereum account detection. The framework accepts account-level Ethereum features, prepares full and reduced feature sets, evaluates XGBoost, Random Forest, Decision Tree and SVM models using stratified crossvalidation, and applies SHAP to interpret the selected model. The pilot dataset is used only to demonstrate data structure and quality checks; final conclusions about model performance will be drawn only after the complete experimental evaluation.

REFERENCES

1. Farrugia, S., Ellul, J., & Azzopardi, G. (2020). *Detection of illicit accounts over the Ethereum blockchain*. Expert Systems with Applications, 150, 113318. DOI: 10.1016/j.eswa.2020.113318.

2. Azzopardi, G., Farrugia, S., & Ellul, J. (2021). *Detection of illicit accounts over the Ethereum blockchain* [Dataset]. University of Groningen. DOI: 10.34894/GKAQYN.

3. Practical 06 instruction sheet supplied for the Research Methodology coursework: "Sample Data Collection and Design the Software Framework."

Submission attachment: An anonymised pilot CSV containing the 15record pilot subset is provided separately with this report.

[Practical No - 7]

Perform Data Analysis / Develop an Algorithm

Aim : To perform data analysis on study habits and academic performance by examining daily study hours, attendance, assignment completion, and examination marks.

Research Problem

A college wants to understand how students' daily study habits relate to their academic performance. The analysis examines whether students who spend more time studying tend to obtain higher examination marks.

Dataset Required

The dataset contains the following fields:

Field	Description
Anonymous Student ID	Unique student identifier
Daily Study Hours	Hours spent studying per day
Attendance Percentage	Percentage of classes attended
Assignment Completion Percentage	Percentage of assignments completed
Examination Marks	Marks obtained out of 100

Dataset and Downloadable Files

The cleaned dataset contains 30 student records. The original sample also includes deliberately introduced data-quality problems so you can document the cleaning process.

Cleaned dataset preview

These are the first 10 records from the complete downloadable dataset.

Student ID	Study hours	Attendance %	Assignments %	Exam marks
S001	0.50	62	60	42
S002	0.75	68	65	48
S003	0.80	72	70	51
S004	0.40	55	50	38
S005	0.90	75	75	55
S006	0.60	65	85*	45
S007	0.25	50	45	35
S008	0.95	78	80	58
S009	1.20	70	70	50
S010	1.50	74	72	57

Task 1 – Data Quality Checking

The raw dataset was checked for missing values, duplicate records, and invalid entries.

Check	Finding	Action taken
Missing values	1 missing assignment value	Filled using the median
Duplicate records	1 duplicate row	Removed duplicate
Invalid entries	S031 had −2 study hours	Removed invalid record
Final records	30	Used for analysis

After cleaning, 30 records remained for the statistical calculations.

Task 2 – Summary Statistics

The mean, median, minimum, and maximum were calculated for study hours and examination marks. The following results come from the simulated sample.

Statistic	Daily study hours	Examination marks
Count	30	30
Mean	2.40	66.13
Median	2.30	69
Minimum	0.25	35
Maximum	5.50	94

Interpretation: The average daily study time was 2.40 hours, while the average examination score was 66.13 out of 100.

Task 3 – Grouping Students

Students were divided into the three groups specified in the reference PDF.

Study-hour group	Number of students
Below 1 hour	8
1–3 hours	12
Above 3 hours	10
Total	30

Task 4 – Compare Average Examination Marks

The average examination marks were calculated for each study-hour group.

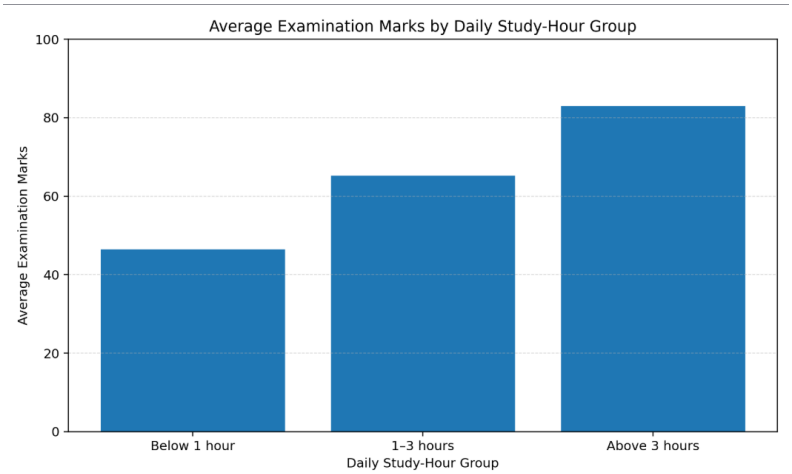
Study-hour group	Students	Average marks
Below 1 hour	8	46.50
1–3 hours	12	65.17
Above 3 hours	10	83.00

Finding: In this simulated dataset, the group studying for more than three hours per day had the highest average examination marks.

9. Task 5 – Charts

Figure 1: Average Examination Marks by Study-Hour Group

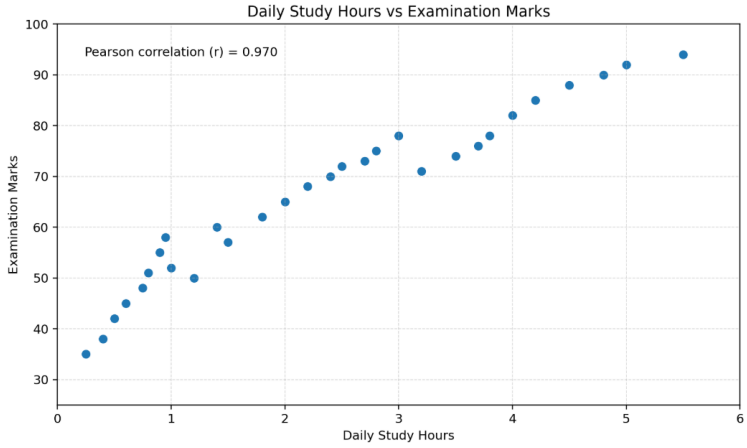
Bar chart showing average examination marks by daily study-hour group



Interpretation: The bar chart shows increasing average marks across the three study-hour groups in this sample.

Figure 2: Daily Study Hours vs Examination Marks

Scatter plot showing daily study hours versus examination marks



The Pearson correlation coefficient was calculated as r=0.970r = 0.970r=0.970 for this simulated dataset.

Interpretation: The scatter plot shows a strong positive relationship between study hours and examination marks in this sample. However, correlation alone does not establish that studying longer causes higher marks.

Task 6 – Conclusion

The analysis examined the relationship between daily study hours and examination marks using 30 simulated student records. After checking and cleaning the data, summary statistics and group averages were calculated, and two charts were generated. Students in the above-three-hour group obtained the highest average marks, while the correlation analysis showed a strong positive relationship between study hours and marks. These results illustrate the analysis procedure; real student data would be needed to draw conclusions about actual students. Other factors, such as attendance, prior knowledge, and assignment completion, may also influence performance.

[Practical No - 8]
Statistical Summaries, Charts and Testing

Study Hours and Academic Performance

Aim : To apply descriptive statistics, visualisation and an appropriate statistical test to answer a research question.

Research Question

Is there a relationship between weekly study hours and examination scores among college students?

Data note: no real dataset was supplied, so a simulated dataset of 80 anonymous students across four programmes (BA, BBA, BCom, BSc) was generated, with a few errors added to practise cleaning. Replace it with real data to get genuine results; the method stays the same.

Data Cleaning

The raw file held 81 records. After cleaning, 77 remained.

Issue found	Action
1 duplicate record	Removed
1 missing examination score	Record removed
2 invalid entries (200 study hours per week; score of -10)	Records removed
Inconsistent programme label ("bcom ")	Standardised to BCom

Descriptive Statistics

Statistic	Weekly study hours	Exam score
Count	77	77
Mean	13.73	58.68
Median	14	60
Mode	15	56, 57, 61, 64 (four tied)
Standard deviation	4.90	11.63
Minimum	3	27
Maximum	26	96

Mean and median are close for both variables, so both are roughly symmetrical. Scores have no single most common value; four scores each appear the same number of times.

Programme-wise Frequency Table

Programme	Frequency	Percent
BA	17	22.1%
BBA	26	33.8%
BCom	25	32.5%
BSc	9	11.7%
Total	77	100%

BBA and BCom together make up two-thirds of the sample, while BSc is the smallest group.

Charts

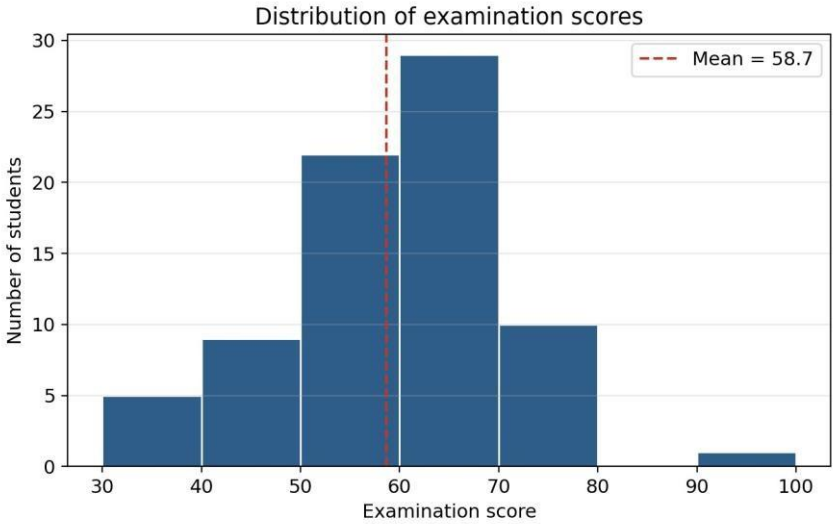


Figure 1: Histogram of examination scores

Scores are centred around 55–65, with fewer students at the very low and very high ends, a roughly bell-shaped pattern (Shapiro-Wilk p = 0.16, so no significant departure from normality).

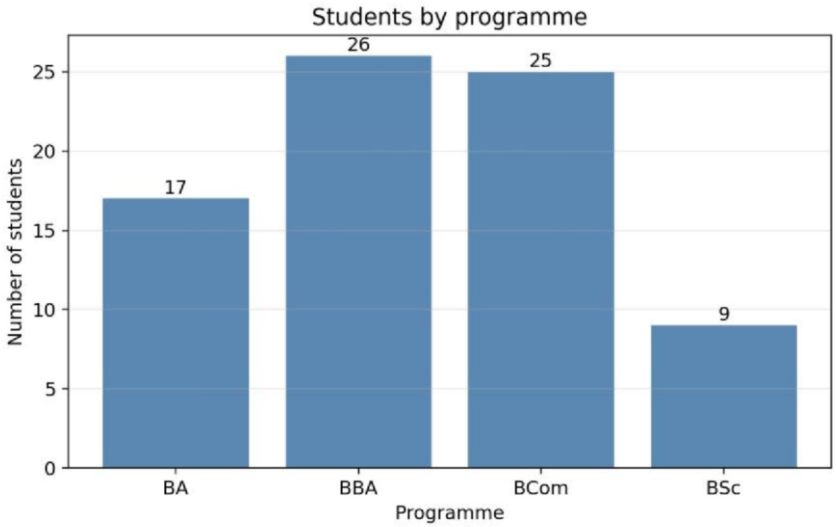


Figure 2: Number of students by programme

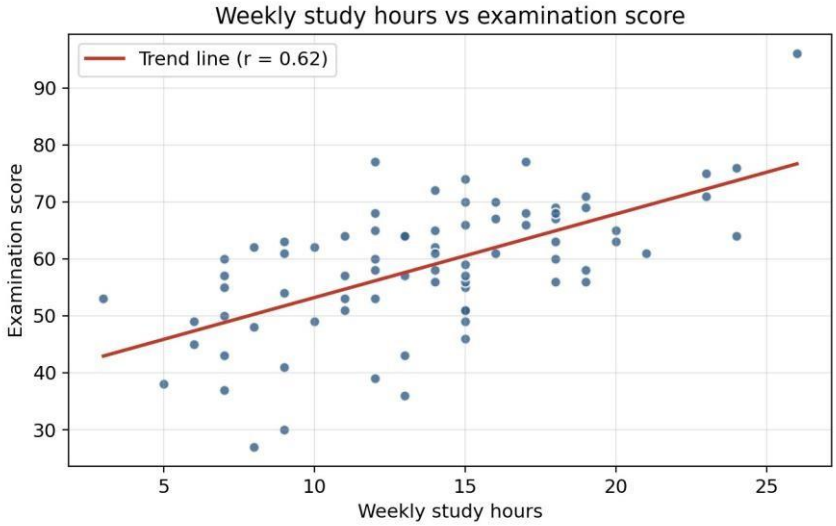


Figure 3: Weekly study hours against examination score

Statistical Test

Because both variables are numeric and the scores look approximately normal, the Pearson correlation test was used. Null hypothesis: there is no linear relationship between weekly study hours and score.

Result	Value
Pearson r	0.62
R-squared	0.38
t (df = 75)	6.81
p-value (two-tailed)	< 0.001
Spearman rho (check)	0.60, p < 0.001
Trend line	about 1.5 extra marks per extra weekly hour

Interpretation

There is a statistically significant, moderately strong positive relationship between weekly study hours and examination scores ($r = 0.62$, $p < 0.001$). Students who study more hours each week tend to score higher, and study hours explain roughly 38% of the variation in scores. The rankbased Spearman test gives almost the same result, so the finding does not depend on the normality assumption.

Important cautions:

- Correlation does not prove causation. Motivation, prior ability or study quality could drive both hours and scores.
- About 62% of the variation in scores is explained by other factors, such as attendance, teaching quality and programme.
- Programme sizes are uneven (BSc has only 9 students), so programme comparisons should be treated cautiously.

Deliverables

- Cleaned dataset, raw dataset and summary tables: Practical08_Dataset_and_Summary.xlsx

Three charts and interpretation: this report